

This lab helped me to understand the different types of machine learning and how they are used for different purposes. The three main types I learned about were supervised, unsupervised, and reinforcement learning. Supervised learning uses labeled data, which means we give the model examples with known answers so it can predict things good. Unsupervised learning is different because the model looks for patterns or groups in data without being given the correct answers. Reinforcement learning involves an agent learning through trial and error by receiving rewards or penalties. Understanding these differences helped me see that machine learning is not one single process and that the type of learning depends on the problem being solved.

One of the most useful parts of the lab was working with the Wine dataset. It had information about all the different types of wine, and we used chemical features like alcohol, malic acid, ash, and alkalinity of ash to predict the wine's class. Before looking through training, I learned why it's important to explore the data and check for things such as missing values and relationships between features. Splitting the data into training and testing sets was also important because it allowed the models to be tested on data they had not already seen. As a result I understood the difference between a model actually learning patterns and simply memorizing the training data.

I also learned how different machine learning models can perform differently on the same dataset. In this lab, I worked with Logistic Regression and a Decision Tree. Logistic Regression performed better on the selected features, reaching an accuracy of 88.9%, while the Decision Tree performed not as good.

I think this means that choosing a model is not just about using the most complicated algorithm. The features being used and how well they relate to the target can have a major effect on the results. I also learned how a confusion matrix and classification report can be used to look at model performance in more detail instead of relying only on accuracy.

Overall, this lab gave me a little bit of a better understanding of the complete machine learning workflow in all its glory, from preparing and exploring data to training and evaluating a model.